TEXAS TECH UNIVERSITY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
CAPSTONE PROJECT LABORATORY

Off-Grid Vehicle Tracker: Device User Guide

Every Screen and Every Button, from Power On to Power Off

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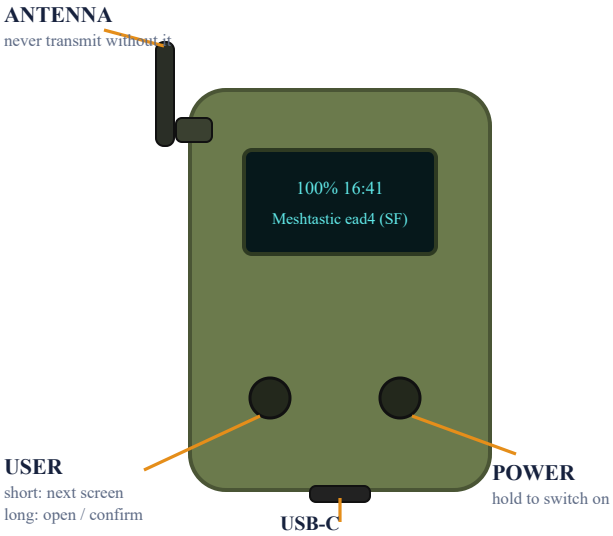
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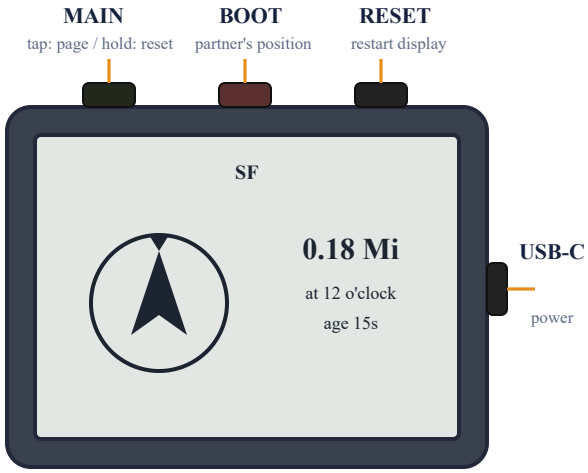
Companion document: the Technical Manual holds the engineering detail, how the system works, and every recovery procedure. This guide is only about operating the devices.

1 The Devices and Their Buttons

The **radio** (green box, antenna) does the talking. The **display** (dashboard screen) shows where your partner is. Four controls run everything.



Radio (Heltec V4). Antenna left, two buttons at the bottom.



Display (Waveshare). Three buttons on top, USB-C on the right.

Control	Short press / tap	Long press / hold
Radio USER	Next screen or menu line	Open action menu; confirm selection
Radio POWER		Switch the radio on
Display MAIN	Next page	Reset tracking; on Messages page, clear history
Display BOOT	Request partner's position now	
Display RESET	Restart the display (rarely needed)	

THE ONE RULE. Never run the radio's battery flat. A dead battery can erase its settings, and the radios stop understanding each other until a laptop fixes it.

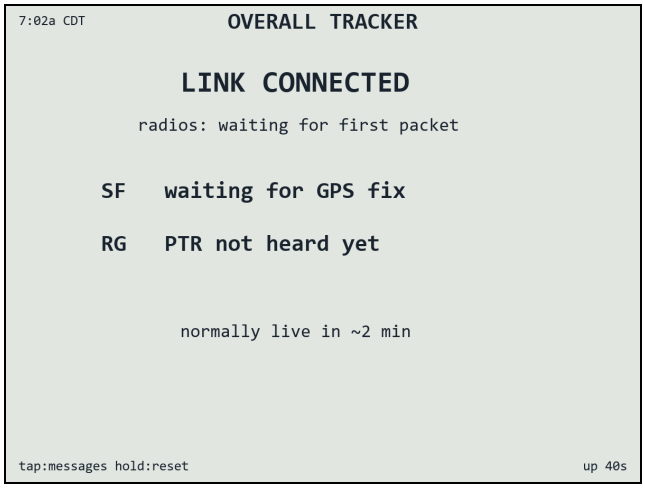
2 Powering On



1. Hold the radio's power button until the screen lights. This home screen is normal. Indoors, "0 online" and "No Sats" are not faults: satellites need sky, and your partner may be off.

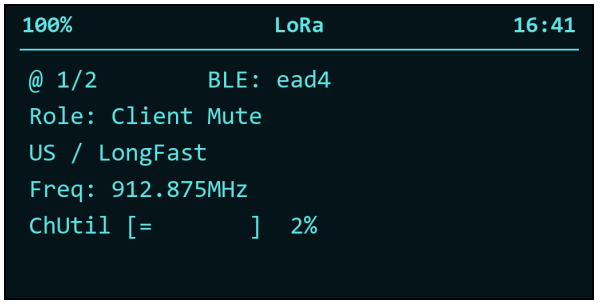


2. Power the display at the same time, like an ignition key. The radio accepts the display only during its first minute, so always switch them on together. Missed it? Power both off and on again.



3. Watch the page fill in. "waiting for GPS fix" clears under open sky; "PTR not heard yet" clears when your partner's radio is first heard. Normally live within two minutes.

3 The Two-Second Pre-Flight Check

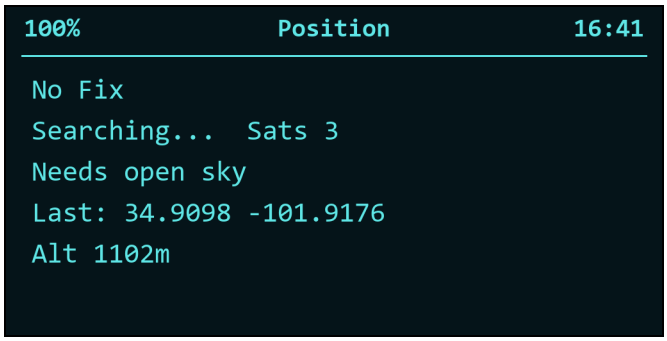


Before every trip, short-press to this **LoRa screen** on both radios and read one number: **Freq: 912.875 MHz**. Both right: everything will work. Either one different: that radio lost its settings and needs the Technical Manual's recovery before the trip.

While you are here, the rest of the screen should read Role: Client Mute and US/LongFast, exactly as shown.

4 The Radio's Screens

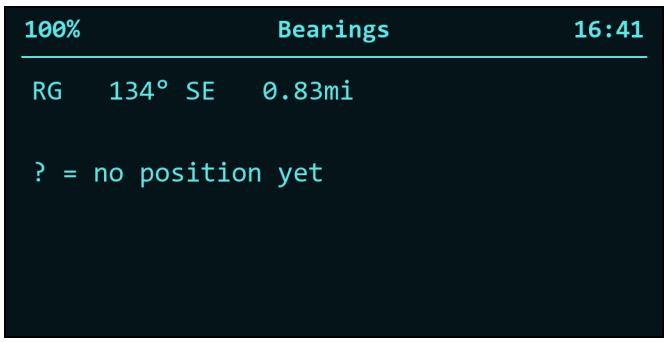
Short presses walk a loop of status screens.



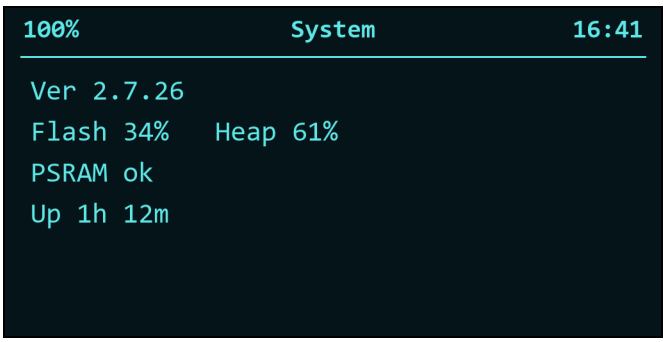
Position. Your own GPS. "No Fix" just means indoors.



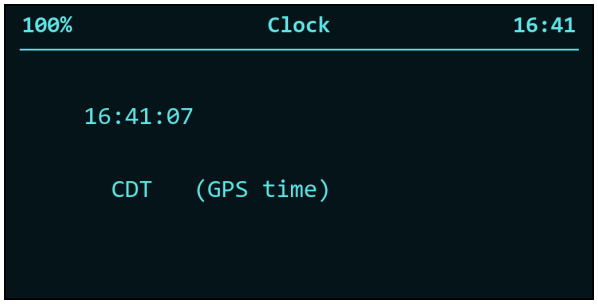
Last Heard. Who was heard, how long ago. The honesty meter.



Bearings. Direction to other radios; "?" means no position yet.



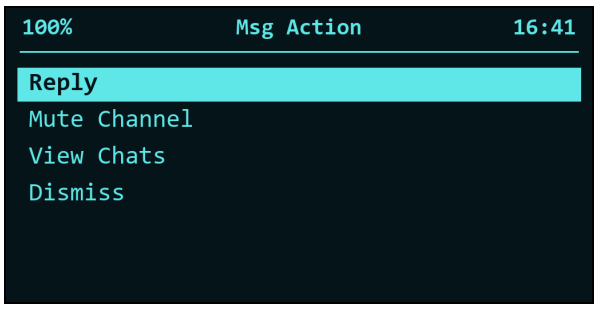
System. Firmware version. Needed only when asked.



Clock. Set by the GPS satellites; nothing to adjust. A correct clock proves the GPS has spoken to this radio.

5 Sending a Message

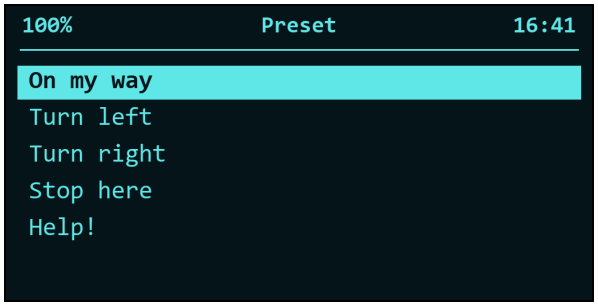
Ready-made messages, so nobody types while driving. Short press moves, long press chooses.



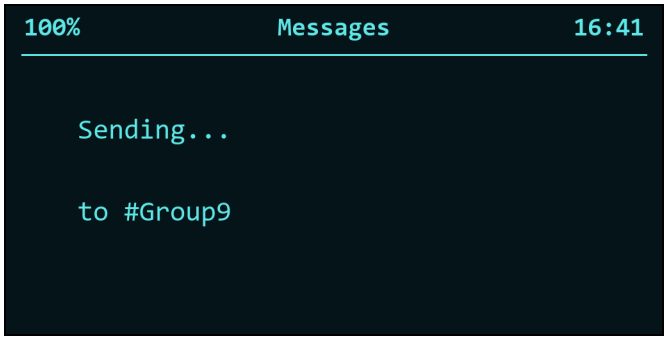
1. Long-press on the chat screen; choose **Reply**.



2. Destination: **#Group9**, the channel both vehicles hear. Skip Free Text.



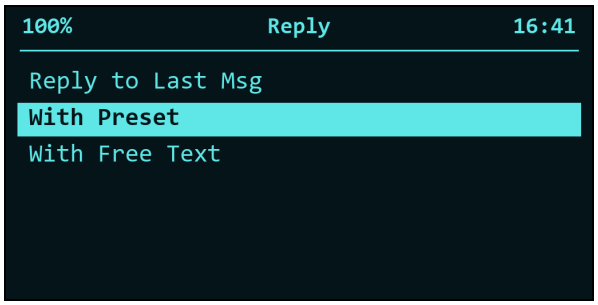
3. Scroll the 21 presets (Help!, Send position, On my way, Turn left, Stop...). Long-press to send.



4. **Sending.** Shown as the message goes out.



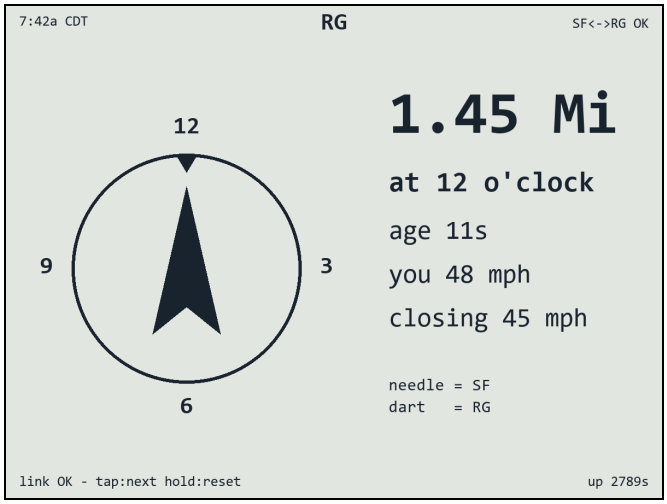
5. **Delivered.** It joins the thread and pops up on the partner's display. An X means not yet confirmed.



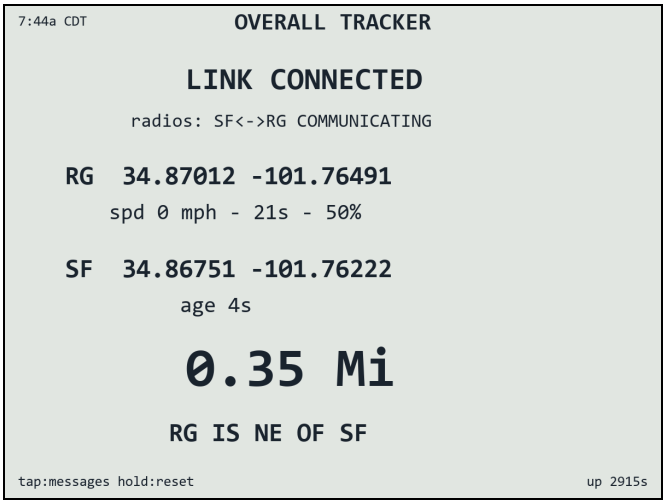
Replying: long-press, **Reply to Last Msg, With Preset**. Every question in the list has its answer in the list.

6 The Display While You Drive

Hands off: it updates itself twice a second. Tap cycles three pages.



Compass. The small marker at the top of the ring is your hood. The dart is your partner: steer until the dart sits under the marker and the partner is dead ahead. The ring is a clock face: 12 is always your hood, 3 your passenger side, 6 behind you, 9 your driver side, so "at 2 o'clock" on the screen and in a message mean the same thing.



Status. Both positions, data ages, partner battery, distance, direction in words.



Messages. Received texts, newest first. Hold here to clear the history.

Partner's position now: press BOOT. The age drops to seconds.

Proven on the road: the screen read 700 ft at 10 o'clock, the driver turned, and the partner's car appeared in the windshield. Trust the clock directions.

7 More Things You Can Do

Every recipe below is the same two moves: short press to reach the line, long press to choose it. All of these are safe.

I want to...	On which device	Do this
Broadcast my position right now	Radio	Long press on the home screen, choose Send Position
Dim the screen to save battery	Radio	Long press on home, choose Sleep Screen; any press wakes it
Read the whole conversation	Radio	Long press on the chat screen, choose View Chats
Silence message pop-ups for a while	Radio	Long press on chat, choose Mute Channel; repeat to unmute
See when the partner was last heard	Radio	Short-press to the Last Heard screen
Check battery and firmware version	Radio	Short-press to the System screen
Switch the clock to 24-hour	Radio	Long press on the clock screen, choose Time Format
Fix the timezone	Radio	Long press on the clock screen, choose Timezone
Flip between display pages	Display	Tap the main button: compass, status, messages
Erase old messages	Display	On the Messages page, hold the main button
Start tracking fresh	Display	On any other page, hold the main button; data rebuilds in seconds
Get the partner's position now	Display	Press BOOT; the age drops to seconds
Restart a confused display	Display	Press RESET; it reconnects by itself

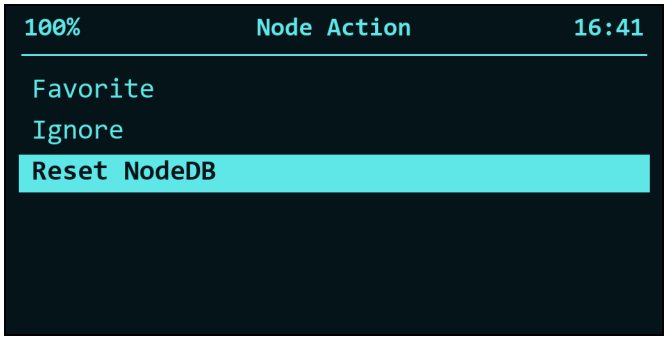
8 How Far It Reaches

Measured on real roads in October 2026, with the antennas inside the cars. The honest rule from all our testing: **distance is not what breaks the link, blocked line of sight is.**

Where you are	What to expect
In town, buildings around	Reliable to about half a mile. Buildings eat radio signal.
Open road, flat country	Fully reliable to 3.26 miles in our test. The screens went stale by 3.75 miles.
Canyon or hill country	The STALE banner comes and goes at any distance, even close together, whenever rock sits between the cars. In our canyon test it appeared inside 1.3 miles, yet updates still slipped through past 4 miles when the land opened up.
Whenever the data is stale	The display shows DATA STALE and the age counts up. Nothing is broken, and no radio can beat rock. Keep driving toward the last known direction; the moment the two cars can "see" each other again it heals by itself, no buttons needed.

WANT MORE RANGE? Mounting the antenna on the roof instead of inside the car can roughly double the distance. Ask the team before changing hardware.

9 Menus to Leave Alone



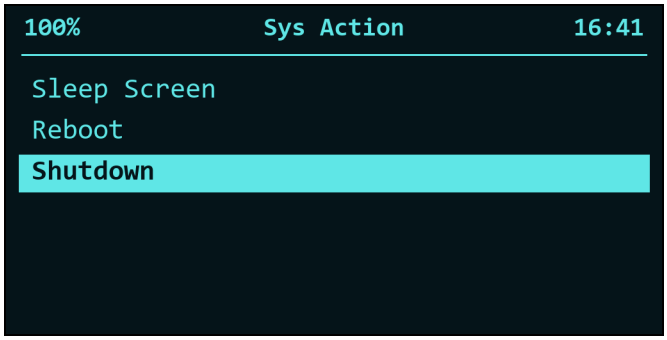
Troubleshooting only: Node Action. Reset NodeDB wipes known radios. Use only when the Technical Manual says so.



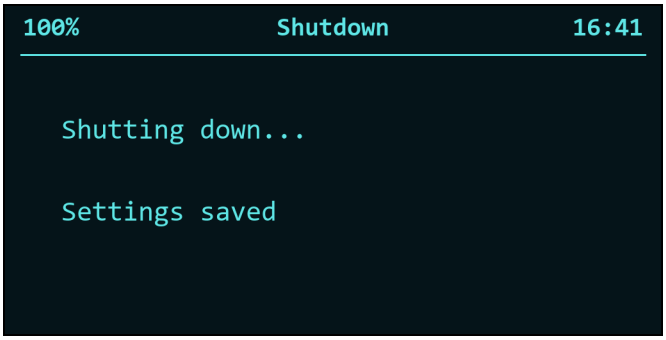
Never change: LoRa Actions. See the warning below.

DO NOT TOUCH LORA ACTIONS. Device Role, Radio Preset, Frequency Slot and LoRa Region are the exact settings that let the two radios hear each other. Changing any of them silences the pair with no error shown. Same for Bluetooth Toggle under System Action: switching it off strands your display.

10 Powering Off and Charging



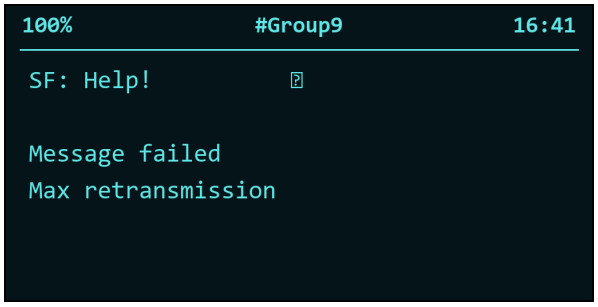
1. Long-press on home, open **System Action**, scroll to Reboot/Shutdown.



2. Choose Shutdown. A clean shutdown always preserves the settings.

Charge over USB-C after every trip. Switched off, the radio holds charge for months. Never store it empty.

11 When Something Looks Wrong



"Message failed": nobody confirmed it; partner off, out of range, or booting. Send again once Last Heard shows them recently.

What you see	Meaning	Do
0 online, No Sats indoors	Normal	Nothing; go outside, partner on.
Frequency is not 912.875, or the radio's name is "Meshtastic_xxxx"	Settings lost (battery ran flat)	Section 12: five minutes with a phone, ninety seconds with the laptop.
Display stuck "SEARCHING" or "CONNECTING"	Missed the pairing window; if it persists, the radio lost its settings	Power both off, then on together. Still stuck after two minutes: Section 12.
DATA STALE while far apart	Out of radio reach	Normal past the ranges in Section 8; heals as you get closer.
Anything else odd	Electronics	Power-cycle both together.

12 Restoring a Radio That Forgot Its Settings

A radio that runs its battery completely flat can lose part of its configuration when it next starts. Nothing is broken and nothing needs re-flashing; the settings only have to be put back. You will know it has happened when one or more of these appear:

Sign	What was lost
LoRa screen reads 906.875 instead of 912.875	The Group9 channel (name and key), so the radio is on the public frequency and cannot hear the partner.
Home screen shows "Meshtastic_ead4" or "Meshtastic_bb5c" instead of SF or RG	The owner name.
Display stays on SEARCHING or CONNECTING for minutes	Bluetooth pairing mode went back to "Random PIN", which the display cannot use.
Partner position updates only every few minutes, or jumps on a coarse grid	Position timing and precision.
The preset message list is the factory one	The canned messages (Section 5).

The losses come in batches, so always restore everything, not just the one thing you noticed. There are two ways to do it. The phone way needs no laptop and works at the roadside; the laptop way is faster and restores every setting at once.

12.1 With a phone, no laptop: about five minutes

You need: the Meshtastic app (free, Android or iOS) and the Group9 channel QR code. Every team member keeps the QR code in their phone's photos; it is also in the team's private hardware notes. The QR code contains the channel key: never post it, print it, or send it outside the team.

1. Switch on the radio by itself. Leave the display off until the end.
2. In the app, open the connection screen (Bluetooth) and tap the radio. It appears as **Meshtastic_ead4** (the SF radio) or **Meshtastic_bb5c** (the RG radio), or as SF or RG if the name survived. If the phone asks for a PIN, read the six digits from the radio's own screen and type them in.
3. **Restore the channel first.** Open the Channels screen, choose the scan-QR option, and scan the Group9 QR code. Accept it. The radio reboots. Check the radio's LoRa screen: it must now read **912.875**. This one step brings back the channel name, the key, the frequency, and full-precision positions.
4. **Bluetooth pairing mode.** Reconnect if the app dropped the radio, then open Radio configuration, Bluetooth, and set Pairing mode to **No PIN**. Save; the radio reboots again. Until this is done the display cannot pair.
5. **Name.** Radio configuration, User: long name and short name both **SF** (or both **RG**).
6. **Role.** Radio configuration, Device: Role **Client Mute**.
7. **Position timing.** Radio configuration, Position: GPS update interval **10 s**; position broadcast interval **60 s**; smart position on, minimum interval **10 s**, minimum distance **20 m**.
8. **Presets, optional.** Module configuration, Canned message: enabled, with the preset list from Section 5. Typed messages work without this step; only the preset list is missing.
9. Disconnect the phone from the radio in the app (the radio accepts one Bluetooth connection at a time). Now power the display: it should show the link within a minute.

> **In a hurry:** steps 3 and 4 are what the partner radio and the display need. Everything else can wait until you are parked.

12.2 With the laptop: about ninety seconds

Each radio has a saved snapshot of its complete, correct configuration: `sf-config.yaml` for the SF radio and `rg-config.yaml` for the RG radio, kept with the team's private hardware notes because they contain the channel key. Plug in the radio alone by USB, with the display unplugged so the radio's port is the only one, and run:

```
python -m meshtastic --configure sf-config.yaml
```

Use `rg-config.yaml` for the RG radio. The radio reboots once or twice while the settings are written. Then check the radio's own screens: the LoRa screen reads **912.875** and the home screen shows **SF** (or **RG**). If anything still reads wrong, run the same command once more; a setting that makes the radio reboot occasionally misses the first pass. If the laptop has more than one serial device attached, add `--port COMx` with the radio's port.

After any deliberate settings change, take a fresh snapshot so the saved file stays correct:

```
python -m meshtastic --export-config > sf-config.yaml
```

12.3 Afterwards

Switch both radios on outdoors. Within a minute, each radio's Last Heard screen should show the other, and the display should show the link. Then do the pre-flight check of Section 3 before driving. To make this section unnecessary: charge after every trip and never store a radio empty (Section 10).

13 Quick Reference

I want to...	Do this
Start up	Antennas on, power radio and display together.
Pre-flight	LoRa screen on both radios reads 912.875.
Send a preset	Long press on chat, #Group9, pick, long press.
Reply fast	Long press, Reply to Last Msg, With Preset.
My position, now	Long press on home, Send Position.
Partner's position, now	BOOT on the display.
Shut down	Long press on home, System Action, Shutdown.
Fix almost anything	Power-cycle both; check 912.875.
Radio forgot its settings	Section 12: scan the Group9 QR, set No PIN; or `--configure` the saved snapshot.

Appendix A Abbreviations and Acronyms

Term	Meaning
BLE	Bluetooth Low Energy; the display-to-radio link. "BLE: ead4" names the paired radio.
BOOT	Display button that requests the partner's position immediately.
CDT	Central Daylight Time, the display's timezone.
ChUtil	Channel utilization; how busy the shared channel is. Single digits is healthy.
Client Mute	The radios' role: talk and listen, never relay. Correct for two vehicles.
Flash / Heap / PSRAM	Radio memory types on the System screen. Never need action.
GPS	Global Positioning System; the satellites all positions come from.
#Group9	The team's private, encrypted channel.
LoRa	Long Range; the radio technique carrying signals for miles on little power.
LongFast	The shared radio speed setting ("US/LongFast").
ME / PTR	Your vehicle / the partner vehicle, on the display's status page.
MHz	Megahertz; a spot on the radio dial. Both radios must show 912.875.
NodeDB	The radio's list of radios it has heard. Reset only for troubleshooting.
OLED	The radio's small glowing screen.
RLCD	Reflective LCD, the dashboard screen; brightest in sunlight.
SF / RG	The two radios' callsigns. Your display shows your partner's.
USB-C	The charging and power plug.
Ver	Firmware version; the radios run Meshtastic 2.7.26.

Off-Grid Vehicle Tracker · Group 9, Texas Tech University ECE Capstone · Screens are exact recreations of the device screens, firmware Meshtastic 2.7.26. Range figures measured on road, October 2026.